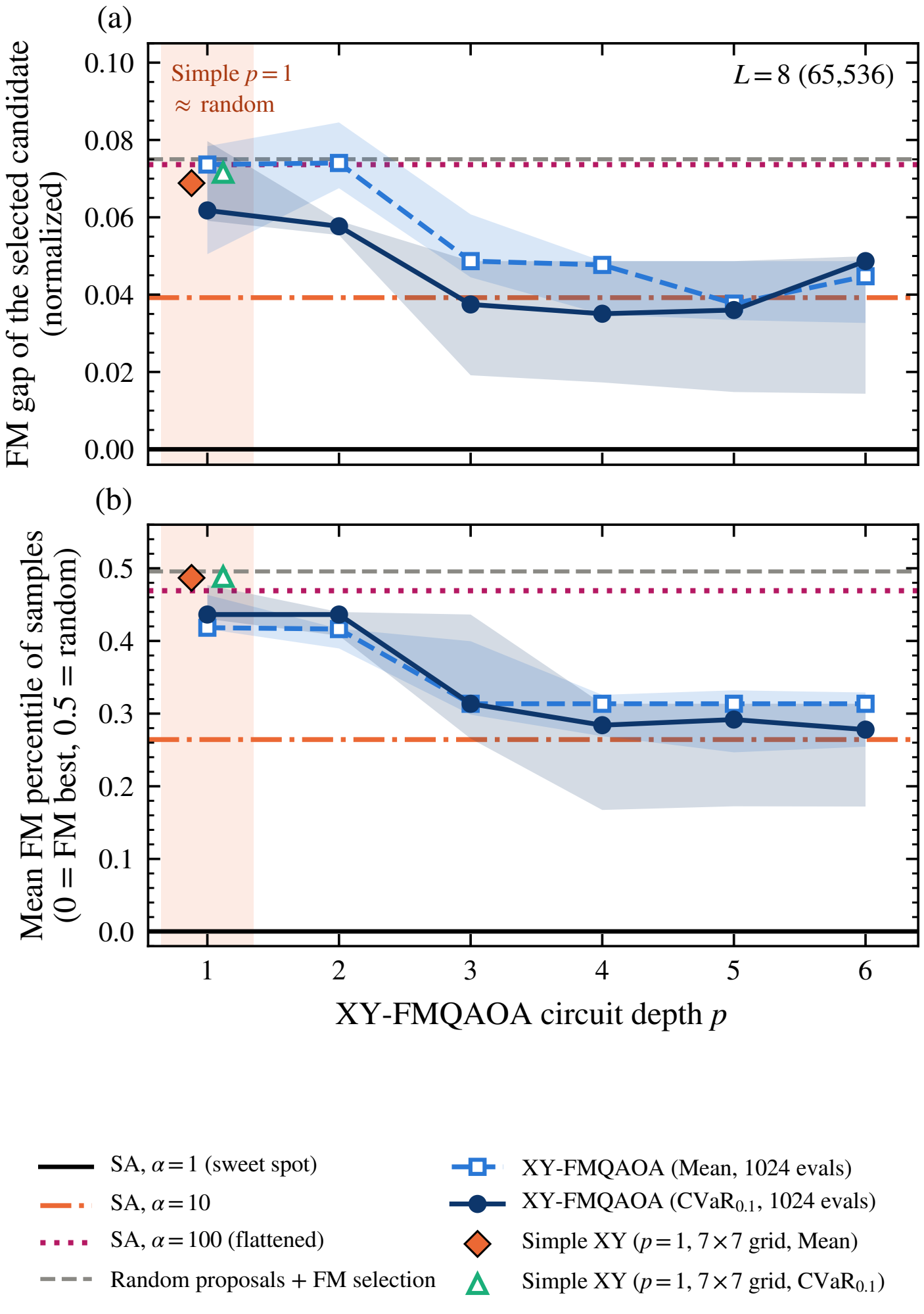
(a)
FM gap of the selected candidate (normalized)
0.10
0.08
0.06
0.04
0.02
0.00
Simple $p=1$ ≈ random
$L=8$ (65,536)
(b)
Mean FM percentile of samples (0 = FM best, 0.5 = random)
0.5
0.4
0.3
0.2
0.1
0.0
1
2
3
4
5
6
XY-FMQAOA circuit depth p
SA, $\alpha=1$ (sweet spot)
SA, $\alpha=10$
SA, $\alpha=100$ (flattened)
Random proposals + FM selection
XY-FMQAOA (Mean, 1024 evals)
XY-FMQAOA ($CVaR_{0.1}$, 1024 evals)
Simple XY ($p=1$, 7×7 grid, Mean)
Simple XY ($p=1$, 7×7 grid, $CVaR_{0.1}$)